

isting cities into smartcities. Majority of the cities in India are over-crowded, and retrofitting these with new technology is a huge task.

According to a survey, India has five out of 20 most-densely-populated cities in the world. Cities like Mumbai and Kolkata are ten times more dense and populated than New York in the USA.

"India presents unique challenges, such as extremely-high population density, congested public spaces, poor penetration of the Internet, maturity of market and old stereotypes in public perception. In such a scenario, we cannot just pick up global solutions and blindly adopt these. We will have to adapt, as opposed to adopt, the technology to meet our special needs," adds Malhan.

Swaminathan believes, "Not one model can fit all. It depends on the physiography and connectivity of the city for successful implementation of smartcity solutions."

Maintaining aesthetics of a city is also a major challenge in front of the government. The government has a tough task of introducing and implementing new technology measures without tempering with the aesthetic values of the city.

"Smartsolutions should be integrated keeping in mind the overall ambience of Indian terrain. There are many tech solutions that are required to be customised according to Indian lifestyles and habits. If not done properly, it might lead to mismanagement and chaos. Particularly, traffic integration is one of the founding pillars of smartcities in India, as majority of the masses is not tech-savvy and lacks awareness. There should be an intense awareness programme for the people to comprehend technology and compel them to adhere to norms," says Saund.

A PWC and NASSCOM consultation report on smartcities says, "The lack of a unified view of the entire city leads to operational inefficiencies and makes cities less resilient during times of emergency due to real-time availability of cohesive information, better

resource allocation and infrastructure management."

Funding is also a major problem in India. The union minister of the Urban Development Ministry of Housing and Urban Poverty Alleviation, government of India, made it clear in a PHD conference, when he said, "Smartcities would need close to 70 trillion of investment and to be executed through PPP mode, for which land will be sourced through banking and financial institutions including multi-lateral agencies."

Apart from policy guidelines and public infrastructure (both of which need to be addressed by respective governments), there are other technical challenges.

Some of the major challenges highlighted by industry experts are:

Infrastructure. It will require evolution of sensors and choosing of the right communication standards, apart from upgradation of existing infrastructure for compatibility and enabling smartcity communication.

Network management. Connectivity solutions available across the city to handle the volume, variety and velocity of data need to be put in place, along with evolution of analytics using Big Data tools to provide citizen-specific insights. The data needs to be correlated and trended for optimum utilisation of various resources

Utilities. There are existing technology challenges for energy storage, renewables, generation and creation of micro-grids, which are still to be addressed.

Smart future

Smartcity as a concept brought into limelight various technology measures, which can be collectively used for enhancing the standards and style of living in a city. The future is significantly brighter if these cities are constructed and developed in a smart manner. The concept that has gained significant amount of importance in developed countries is gradually gaining traction in developing countries like ours. ●



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